

Q1. Write a Python program to create an employee_data DataFrame with the attributes: Emp_ID, Age, Dept, Years_Experience, and Performance_Rating. Perform data cleaning by handling missing values, remove duplicate records and detecting outliers using the IQR method.

[10 Marks]

Q2. Consider the World Cities dataset containing attributes - city, country, latitude, longitude, and population information. Perform the following tasks:

- a. Create a Geospatial Visualization showing the locations of cities on a world map, using marker size to represent population.
- b. Create a Treemap to visualize the population distribution of cities.

[20 Marks]

OR

Q2. Use the Crop Yield dataset to apply Simple Linear Regression for predicting crop yield from rainfall. Handle missing values, split the data into training and testing sets, and evaluate the model using MAE and R^2 Score. Create a Scatter Plot with a Regression Line to visualize the relationship.

[20 Marks]

Q3. Viva

[5 Marks]